

1. Executive Summary

my-job-agent is an autonomous personal automation service for Swarna Sekhar Dhar with two major subsystems sharing one NestJS + TypeORM + MySQL (OCI MDS) codebase on branch dev:

A) JOB-APPLICATION AGENT — scouts jobs from 12+ sources (Naukri, RemoteOK, Remotive, LinkedIn, Norway, H1B-US, A1 Group, TinyFish, finn.no...), tailors an ATS PDF CV + cover letter per job description, applies through the JD-stated process (company ATS → HR email → portal easy-apply last), and tracks both-side history. Nothing sends without user approval: an hourly scout → prepare loop builds a pre-apply review queue, the user approves items, and a muhurta (Vedic astrology) sweep sends everything together inside shubh windows. 170 leads captured; 13 applications submitted via real send paths; IMAP reply tracking polls two mailboxes every 30 minutes.

B) PAPER F&O; OPTION TRADING DESK — an autonomous paper-trading engine for Indian index option chains (NIFTY 50, BANKNIFTY; SENSEX/FINNIFTY by design). It streams LIVE market data (FYERS WebSocket primary, Yahoo fallback), registers real option contracts, runs a candidate-ranking DECISION ENGINE over the whole registered universe with a ■5,000 capital filter, and opens paper positions only on CE/PE option contracts — never on the raw index. Session-scoped tick archival keeps live tables today-only and every tick in history. (Full trading report: OPTION_DESK_STATUS_2026-09-04.pdf, same folder.)

This whole-project report adds the job-application side, shared infrastructure, the integrated feature TODO plan, and cross-cutting gaps. Both subsystems are at an honest early stage: the plumbing is real and verified end-to-end, but the job side is capped by browser-automation gates and a user-driven approval loop, and the trading signal layer is still heuristic (no ML, no backtest, no labeled history yet).

2. Project Overview

2.1 Purpose & scope

Purpose (verbatim from PROJECT_PROMPT.md): "Autonomous job-search agent for Swarna Sekhar Dhar — scouts jobs, tailors ATS PDF CV per job description, applies directly (JD-stated process → company ATS → HR email → portal easy-apply last), tracks both-side history, self-improves from outcomes." The trading desk was added (2026-09-01+) as a paper-only F&O; research arm with its own discipline (■5k envelope, observation sessions, decay-aware signals, astro gating).

2.2 Stack & topology

ORM, Node on pm2 — local dev server (port 3010, dashboard /, Swagger /docs)
DS (10.0.0.99:3306), reachable via SSH tunnel through Dhargent (127.0.0.1:3307); schema myjob_agent, 21 tables
in via Cloudflare tunnel (cloudflared, not yet reboot-surviving)
model: login required for local AND public access; ConditionalAuthGuard with no loopback exemption (tunnel arrives as loopback); @E
10.60.10 — headless "trading-agent" (pm2), standalone module, shares the same MDS DB
on/my-job-agent, branch dev — 107 commits; conventional commits; .env tracked per convention (no secrets in code)

2.3 Module map (src/)

ers (Naukri, RemoteOK, Remotive, LinkedIn, Norway, H1B, A1Group, TinyFish, finn, Monster/foundit) + hourly fetch loop + lead scoring
ecycle: prepare (CV/cover-letter/email), pre-apply review queue, approve/hold, send sweeps, tracking calendar + failed-page + day-list
ta engine (Lahiri ayanamsa, panchanga, tithi/nakshatra/weekday scoring), job astro-match scoring, send-window sweep
ssword login, ConditionalAuthGuard, rate limits, change/forgot password
ctice bank + page
/dedupe (never re-apply same lead or portal external id)
ofile + ATS PDF CV builder (pdfkit), geo/employer CV rules
side-income opportunities + page
SMTP/IMAP), RetryBackoff, learning-weights stats, common utils
portfolios/trades/ledger, market-data feed (FYERS WS + Yahoo fallback), option contracts+quotes, decay engine, candidate-ranking dec
headless driver (Dhargent): 10s session tick 09:15–15:30 IST, position-driven exits, tick archival

3. Subsystem A — Job-Application Agent

3.1 Pipeline (verified working)

SCOUT (hourly, portal cap per run, dedupe) → SCORE (title veto: designer/QA/sales never match; title-hit-or-3-skills; 40% floor) → PREPARE (tailored ATS PDF CV + cover letter + email draft; FR-19 channel detection: JD-stated process → company ATS → HR email → portal easy-apply last) → USER REVIEW (/pre-apply-page: approve / hold / pause / upload corrected CV) → MUHURTA SWEEP (every 10 min inside a shubh window sends ALL approved items together) → TRACK (statuses queued → sent; reply polling both mailboxes every 30 min, noise-filtered; status_updates stores content+date).

3.2 Live state (2026-09-04)

nyfish 18, remoteok 14, h1b 8, norway 6, remotive 6, linkedin 4, a1group 2, bicsom/bihartechgroup/finn/zebra 1 each
bmitted 13, failed 4 (Remotive/Lemon.io Cloudflare), needs_info 1 (LinkedIn Continental — not visa-sponsored)
iting user approval incl. ZEBRA Techies → hr@zebratechies.com), hold 6, sent 13, failed 22 (channel-less naukri rows; collapsed in UI,
· 08-31: 1 · 09-03: 3 (latest)
p passwords) — Gmail primary sender + OTP reader + recruiter reply polling
EST_HOUR to bapay.9@gmail.com

3.3 Verified capabilities & known caps

Real sends verified end-to-end: Naukri apply workflow (12+ real auto-applications), BiCSOM via Fluent Forms (API-driven, no browser popup), ZEBRA Techies prepared (awaiting approval), portal easy-apply + company email paths. Learning weights live (channel/portal/hour stats → bestChannel recommendation). Browser automation (FR-15 playwright-core + system Chrome) is VERIFIED but currently BLOCKED at runtime: the Chrome profile has no remote-debugging flag and cua_browser_prepare requires user approval — all browser-gated backlog (Remotive retry, 8 H1B leads, LinkedIn needs_info) waits on that single user decision.

4. Subsystem B — Paper F&O; Option Trading Desk

Full detail in OPTION_DESK_STATUS_2026-09-04.pdf (same folder). Summary of the live architecture:

4.1 Data → Decision → Execution (implemented)

market_snapshots (index) + fnf_option_quotes (premiums/bid-ask/vol/OI/Greeks) with *_history archives; session archival at 15:30 + data from FNO_MARKET_DATA_SYMBOLS at boot
+ data-age); muhurta/astro window gating
reject premium×lot×lots > headroom (■5k filter) → liquidity/spread/freshness/expiry gates → weighted score (strategy fit 40, liquidity 25, stored target/stop; premium P&L (exit-entry)×units; FYERS per-segment costs (options ■20/order, STT 0.05% sell, NSE txn, stamp, S ■5,000 = capital constraint, NOT instrument-selection rule; NO TRADE if nothing qualifies; buy-side long options only (bounded loss)

4.2 Trading live state (2026-09-04 22:18 IST)

Metric	Value
Portfolio	sandbox-live — capital ■5,000.00, deployed ■0.00, netPnl −■200.74, ceiling ■4,799.26, totalCost ■64.54, autoTrade O
Closed trades	2 legacy index BUYs (scope-drift from T-03, closed per user instruction)
Open trades	0 — desk ready for next session (Mon 09-07 09:15 IST)
Feed	FYERS primary live, 35 symbols subscribed (Dhargent pid 30206 on newest build)
Kit	11f773e — candidate-ranking DECISION ENGINE + 32-contract universe, pushed origin/dev

4.3 Milestones already shipped

2026-08-31 F&O; options page + FYERS v3 adapter · 09-01 market-data inspector + Yahoo safeguard + paper rules doc · 09-02 operator auth model · 09-03 ■5k envelope + decay engine + UI round 2 (failed-page 500, calendar clicks, pre-apply failed cards) · 09-04 T-03 observe session (2 legacy index opens, none hit target/stop, held to close), FYERS token refresh (user click), session tick archival, T-04 option-chain desk (index=reference-only hard guard), candidate-ranking DECISION ENGINE, universe 40 contracts.

5. Shared Infrastructure

login required local + public; no loopback bypass; SESSION_SECRET rotated to random 64-hex (commit 35000f9)
Lahiri ayanamsa, panchanga cross-checked vs published panchang (2026-08-27), shubh-window send sweep
14 calibration rows; day-wise rectification from closed-trade outcomes (winners ease rate 0.15, losers tighten; window drift)
both mailboxes every 30 min, noise-filtered, encrypted credentials
it NOT reboot-surviving (I-02: systemd unit pending)
lev pm2 my-job-agent; Dhargent trading-agent rsync-deployed from dist/ (no git on host)
ECT_PROMPT.md + docs/* (source of truth), agent_todo_log table, skill playbooks (market-data-paper-trading, option-paper-trading, job)

6. Design Conflicts & Decisions (history as evidence)

Resolution
User: index must be underlying/reference ONLY. Hard guard added; 2 legacy index positions closed on instruction (net -█200.74 incl. 100k was my own "notional headroom" rationalization. Corrected to ceiling = █5,000 = capital, then ceiling = capital + netPnl (auto-grow
User: SENSEX was an EXAMPLE, not a fallback; █5,000 is a capital filter across the whole eligible universe. Docs + engine rewritten t
Chrome profile lacks remote-debugging; cua approval timeouts. Whole browser-gated backlog (H1B×8, Remoteive retry, LinkedIn needs
FYERS access token single-day; expired at T-03 open (-15). Yahoo fallback sanctioned; user login click refreshed token; no auto-refre
Deliberate design signature (user belief system) — shubh-window send sweep + trade gating; acknowledged as statistically unvalidate
snapshots.ts IST-naive; trades UTC. Archive logic compares IST boundaries in the same naive frame; documented to prevent regressi
New entity tables must be created via SQL before boot on prod (gotcha logged; history tables auto-created on dev via synchronize).

7. Feature TODO Plan (do-do list)

Source: PROJECT_PROMPT.md "TODO next session" + agent_todo_log open rows + the trading target architecture. IDs match agent_todo_log / backlog. Prioritized:

7.1 Job-application agent

redirect apply handling (FR-19 always-company, slice 2): detect redirect to company careers page, drive form, login/register handling
automation — the TRUE fix for 22 channel-less naukri failed rows (UI collapses them today; no re-approve without a channel)
by geo-location + employer preference (FR-4)
stion page: free-text Q&A instead of canned answers
fresh while a portal apply run is in flight
unicode sanitization (CV builder + workHistoryJson)
uccess button on main calendar day-list (failed apps shown)
careers adapter (WordPress REST; vacancy list client-side — read block JS)
H1B-US company career pages (Stripe, Anthropic, Klaviyo, Lyft, Coinbase, Adyen, Databricks, Airbnb)
Remote/Lemon.io senior role (failed ×3 Cloudflare) via real browser page or company Greenhouse/Lever
ental needs_info (5 fields; Bengaluru, NOT visa-sponsored — low priority)
ssion-cookie capture (captcha verdict: no bypass); scrape side already live
le HTML → POST /linkedin/refresh for keyword tuning
send prepared application (hr@zebratechies.com, astro 80/100)
982-vs-1981 birth-date flag (PAN + 10th cert vs chart) — affects astrology

7.2 Trading desk (in target-architecture order)

	Why / gate
gin/max-loss validation hardening (bid-ask liquidity partly live in gates)	decision-engine foll
ne: per-instrument rates, expiry-day handling, auto-trade gating on decayed confidence	needs live outcome
/WAP/ATR/gap/ORB/microprice + IV rank/percentile, term structure, OI-delta, P/C ratio	raw tick data exists
X + FINNIFTY chains (quotes probed live; liquidity cut today)	widen the universe
rend/range/high-vol/panic tags per session	feeds scoring
session P&L limit, consecutive-loss pause, volatility shutdown; overnight/multi-session position policy	current envelope is
collection → XGBoost/LightGBM signal engine (label design: target/stop outcome, theta-adjusted premium P&L)	needs hundreds of
o-refresh (remove the daily manual login click)	single-day access t
harness over archived ticks + decision engine	data accumulates c
p: per-strategy/strike/expiry attribution, model comparison, validation gate → live-shadow path	target architecture
ser (Telegram bot exists: @Dharmeas_task_bot)	headless today
g (Zerodha Kite / Angel One) — REAL MONEY, gated on user + paper track record	T-06, far out

7.3 Infrastructure

ID	Item	Gate
I-02	Cloudflare tunnel systemd unit (reboot-survival)	none — code
—	Cron/watchdog health checks for local + Dhargent processes	none — code
—	Monitoring of reply-polling + digest jobs	none — code

8. Cross-Cutting Gaps & Risks (honest)

	Impact
ever)	No ML, no decay-model validation, no backtest — every rule weight is reasoned, not fitted
	13 submitted applications in ~2 weeks; browser gates stall the biggest channels; funnel needs more volume to learn (learning
	Deep-OTM = high-theta lottery tickets at NIFTY $\approx 23,900$ (ATM lot $\approx \blacksquare 13\text{--}26\text{k}$); envelope and strategy conflict at current index
	No option-aware signal: IV, theta decay, event risk not modeled
	FYERS daily token refresh needs a user click; Chrome flags need user approval; tunnel not reboot-safe
	May help or hurt both subsystems; no statistical evidence yet
	All knowledge in PROJECT_PROMPT.md/docs + skills; no tests around the newest trading code (only market-data filter tests)
	Credentials in tracked .env (project convention); operator-password auth with no loopback bypass; secrets never in chat (mas

9. Review Questions for External Advisors

1. Given the approval-gated funnel (scout → prepare → user approve → muhurta sweep), what is the best next lever to raise application VOLUME and QUALITY without burning the user's review time?
2. For the trading desk: is the candidate-ranking scoring model sane, and where should the ■5k envelope trade — deep-OTM single contracts, smaller-lot chains, or should the envelope itself be reconsidered?
3. What is the minimum realistic paper-trade count before XGBoost/LightGBM becomes meaningful, and what label design would you use?
4. Which feature-engine priorities would you pick (IV rank, term structure, OI-delta, P/C ratio, VWAP deviation) given only tick OHLC + premium + OI today?
5. Is single-position-per-signal correct for ■5k, or should concurrent uncorrelated positions be allowed when each fits the envelope?
6. Which mandatory gates would you add to the option candidate filter (IV-percentile floor, min OI, max absolute spread ■, DTE minimum 7...)?
7. How should the muhurta/astro windows be tested honestly alongside the quant signal without confounding either?
8. What is the right overnight/multi-session policy for paper options when the driver only runs 09:15–15:30 IST?
9. Where would you add automated tests first in this codebase (the trading decision engine, the decay rectifier, the send-sweep, the CV builder)?
10. Would you prioritize the browser-automation unlock (H1B + Remote channels) over the trading ML pipeline, and why?

Appendix A — Live State Snapshot (2026-09-04 22:2x IST)

Area	State
Git (dev)	11f773e pushed origin/dev; 107 commits; clean tree
Local app	pm2 my-job-agent online (pid 68607, port 3010)
Dhargent	trading-agent online (newest build), FYERS 35 symbols, driver armed Mon–Fri 09:15–15:30 IST
Trading portfolio	■5,000 capital / ■4,799.26 ceiling / netPnl –■200.74 / deployed ■0 / 40 contracts
Job funnel	170 leads · 44 applications (13 submitted) · pre-apply: 3 ready / 6 hold / 22 failed(no channel)
DB	21 tables in myjob_agent (OCI MDS via tunnel 127.0.0.1:3307)

— End of report —